# Referee Report 2

August 4, 2026

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**Final Recommendation**

**Major Revision**

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## Overall Assessment

The manuscript proposes an original approach to the Navier–Stokes equations based on a new fixed point theorem. The combination of weighted Sobolev spaces, the heat kernel, and the new fixed point argument is mathematically interesting and potentially significant.

After considering the author's explanations and the supplementary material, I no longer regard the manuscript as containing an obvious fatal mathematical error. Several concerns that initially appeared serious—including the contraction property, the new fixed point theorem, Lemma 3, parts of Lemma 5, the analyticity argument, and the energy inequality—have plausible resolutions.

Nevertheless, the present manuscript is not yet self-contained. Essential arguments are distributed between the main text, later corrections, supplementary notes, and explanatory figures. A reader should not have to reconstruct the final proof from multiple revisions.

For publication, the manuscript should be rewritten so that every theorem and proof appears once, in its final form, within the main text.

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## Strengths

- Original fixed point theorem.
- Interesting application to the Navier–Stokes equations.

- Careful construction of weighted Sobolev spaces.
- Ambitious attempt to obtain smoothness and analyticity from a unified functional framework.
- Most mathematical objections raised during the review appear to have reasonable answers.

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## Required Revisions

1. Incorporate every supplementary proof into the manuscript.
2. Remove all provisional material ("Correction", "Counterexample", "Conjecture", "We expect", etc.).
3. Rewrite Lemma 3 and Lemma 5 so that the final proofs appear only once.
4. Clarify the proof of the energy inequality in its final form.
5. Improve the organization of the Main Theorem.
6. Clarify the notation and definitions so that no reference to supplementary explanations is necessary.

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## Final Evaluation

| **Category** | **Evaluation** |
|---|---|
| Originality | Excellent |
| Mathematical Interest | Excellent |
| Technical Quality | Good |
| Exposition | Fair |
| Recommendation | Major Revision |